

Gender Differences in Life-Course Theory of Recidivism: A Survival Analysis

Statistical Consulting Presentation

Based on a study by Brent B. Benda

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Benda, B. B. (2005). Gender Differences in Life-Course Theory of Recidivism: A Survival Analysis. *International Journal of Offender Therapy and Comparative Criminology*, 49(3), 325-342.

Outline

- **Key Concepts**
- **Problem**
- **Motivation**
 - Study background and research objectives
 - Role of social bonding and adverse experiences
(Why timing and life-course transitions matter)
- **Data Description**
- **Method: Cox Regression Model**
 - Model overview, assumptions, and hazard ratios
- **Results**
- **Data Analysis Highlights and Interpretation**
- **Conclusion and Limitations**

Key Concepts

- **Recidivism**

- The tendency of a convicted criminal to reoffend, such as returning to prison after prior release or violating parole

- **Life-Course Theory | Sampson & Laub**

- Desistance from crime can be explained by social bonding that occurs in adulthood
- As social bonds grow, the incentive for avoiding crime increases due to more being at stake

- **Cox Regression**

- A model used in survival analysis to estimate how predictors affect the survival time of an event
- Shows the risk of an event occurring based on each variable, using hazard ratios (HRs)

Problem

- **Approximately 83%** of state prisoners released in 2005 were rearrested within **9 years**, with **44%** arrested within the first year ([Bureau of Justice Statistics, 2018](#))
- **High recidivism rates** remain a persistent issue in the U.S. among individuals released from correctional programs like boot camps
- **Gender-specific factors** influencing when individuals reoffend are underexplored

Motivation

- **Three Main Purposes**

- **Predictive Variables:**

- Examine how well certain variables allow us to predict recidivism for men and women graduates of a boot camp during a 5-year follow-up period.

- **Social Bonding Mediation:**

- Analyze if these social bonding factors mediate the effects of abuse, adverse feelings and thoughts on recidivism

- **Emotional Change Impact:**

- Determine if changes in adverse feelings and thoughts during the boot camp program can predict recidivism

Data Description

- **Sample Size**

- 600 bootcamp graduates from a Midwestern U.S. state
- 300 women & 300 men
- First time referral, sentenced ≤ 10 years, no recorded violence in state system, $IQ \geq 70$, no physical and psychological problems (including drug addiction), and ≥ 20 years old

- **Variables Measured**

- **Predictors:** Age, age at first arrest, race, urban/rural residence, childhood SA, childhood physical abuse, recent SA, recent physical abuse, stress, depression, aggression, fearfulness, suicidal thoughts, gang membership, carrying weapons, selling drugs, drug use, alcohol abuse, criminal peer pressure, criminal partner
- **Life-Course Transitions:** number of children, years of education, job satisfaction, partner relationships, friendships, family relationships

Methods

- **Data Collection**

- Pre-test, post-test, and follow up questionnaires (likert scale)
- 12 men (4%) had missing recidivism data (missing not censored), remove from analysis

- **Cox Proportional Hazard Model:** $h(t|z_1, z_2, \dots, z_m) = h_0(t) \cdot \exp(\beta_1 z_1 + \dots + \beta_m z_m)$

- **Analyze time to event (recidivism)**, uses hazard ratio (HR) for every predictor
- $HR > 1$ (increase in recidivism) & $HR < 1$ (decrease in recidivism)
- First hierarchy including backgrounds and psychological risk factors
- Second hierarchy adds life-course transitions
- Chosen for its ability to handle censored data and time-to-event outcomes

Results

COX'S PROPORTIONAL HAZARDS MODEL OF RECIDIVISM FOR WOMEN

	First Hierarchy		Second Hierarchy	
	Beta	HR	Beta	HR
Age	-0.33**	0.72	-0.39**	0.68
Age of First Arrest	-0.15*	0.86	-0.06	0.94
Race (0 = White)	0.06	1.06	0.04	1.04
Residence (0 = Rural)	0.47**	1.60	0.48**	1.62
Childhood Sexual Abuse	0.63**	1.88	0.57**	1.77
Childhood Physical Abuse	0.58*	1.79	0.45**	1.56
Recent Sexual Abuse	0.80**	2.23	0.77**	2.17
Recent Physical Abuse	0.70**	2.02	0.67**	1.96
Adverse Feelings and Thoughts				
Stress	0.77**	2.15	0.74**	2.10
Depression	0.63**	1.88	0.59**	1.80
Aggression	0.09	1.09	0.06	1.06
Fearfulness	0.54**	1.73	0.51**	1.66
Suicidal Thoughts	0.60**	1.82	0.49**	1.63
Two Months After Graduation				
Criminal Partner (0 = No)	0.64**	1.90	0.60**	1.82
Criminal Peer Associations	0.22**	1.25	0.19*	1.21
Carry Weapons	0.05	1.05	0.04	1.04
Alcohol Abuse	0.12*	1.13	0.04	1.04
Drug Use	0.45**	1.56	0.42**	1.52
Selling Drugs	0.29**	1.34	0.27**	1.31
Life Transition Factors				
Number of Children			-1.24**	0.29
Years of Education			-0.07	0.93
Job Satisfaction			-0.21*	0.81
Partner Relationships			-0.65**	0.52
Friendships			-0.36**	0.70
Family Relations			-0.43**	0.65
-2 log likelihood	221.93**		301.73**	

COX'S PROPORTIONAL HAZARDS MODEL OF RECIDIVISM FOR MEN

	First Hierarchy		Second Hierarchy	
	Beta	HR	Beta	HR
Age	-0.46**	0.63	-0.43**	0.65
Age of First Arrest	-0.51**	0.60	-0.42**	0.66
Race (0 = White)	0.03	1.03	0.01	1.01
Residence (0 = Rural)	0.12*	1.13	0.10*	1.10
Childhood Sexual Abuse	0.35**	1.42	0.32**	1.37
Childhood Physical Abuse	0.31*	1.36	0.28**	1.31
Recent Sexual Abuse	0.01	1.01	0.01	1.01
Recent Physical Abuse	0.06	1.06	0.03	1.03
Adverse Feelings and Thoughts				
Stress	0.37**	1.44	0.34**	1.40
Depression	0.09	1.09	0.07	1.07
Aggression	0.54**	1.72	0.53**	1.70
Fearfulness	0.18*	1.20	0.09	1.09
Suicidal Thoughts	0.17*	1.18	0.14*	1.15
Two Months After Graduation				
Criminal Partner (0 = No)	0.11*	1.12	0.07	1.07
Criminal Peer Associations	0.57**	1.77	0.55**	1.73
Carry Weapons	0.50**	1.66	0.48**	1.62
Alcohol Abuse	0.27**	1.32	0.22**	1.25
Drug Use	0.41**	1.50	0.39**	1.48
Selling Drugs	0.23**	1.26	0.18*	1.20
Life Transition Factors				
Number of Children			-0.09	0.91
Years of Education			-0.17*	0.84
Job Satisfaction			-0.69**	0.50
Partner Relationships			-0.26**	0.77
Friendships			-0.06	0.94
Family Relations			-0.07	0.93
-2 log likelihood	210.71**		289.42**	

Highlights

- **Recidivism Rates (Hazard Ratio)**

- **Women: 42.3% (127 women reoffended)**

- **Women risk factor** | recent sexual (2.17) and physical abuse (1.96), and stress (2.10)

- **Women protective factor** | number of children (0.29), partner relations (0.52), and family relations (0.65)

- **Men: 62% (178 men reoffended)**

- **Men risk factor** | criminal peer association (1.73), aggression (1.70) and carrying a weapon (1.62)

- **Men protective factor** | job satisfaction (0.50), partner relations (0.77) and years of education (0.84)

- **Life Course Theory**

- Overall, adult bonds (life-course transitions) do help adults from reoffending

Conclusions / Limitations

- **Recidivism occurs due to different factors affecting males and females**
 - Emotional and abuse related factors risk more for women
 - Behavioral and peer-related factors risk more for men
- **Sample Limitations**
 - Sample only comes from 1 bootcamp
 - Self-reported data
 - Missing data
 - Follow up is only 5 years
- **Future Research**
 - Replicate study in multiple bootcamps
 - Extend follow up beyond 5 years
 - Add trauma-informed programs

References

- Benda, B. B. (2005). Gender Differences in Life-Course Theory of Recidivism: A Survival Analysis. *International Journal of Offender Therapy and Comparative Criminology*, 49(3), 325-342.
- Laub, J. H., & Sampson, R. J. (2003). *Shared beginnings, divergent lives: Delinquent boys to age 70*. Cambridge, MA: Harvard University Press.
- DATAtab Team (2025). DATAtab: Online Statistics Calculator. DATAtab e.U. Graz, Austria.
- Alper, M. E., Durose, M. R., & Markman, J. (2018). 2018 update on prisoner recidivism: A 9-year follow-up period (2005–2014). Bureau of Justice Statistics.

Appendix

CHARACTERISTICS OF THE SAMPLE (N = 600)				
	Men (n = 300)		Women (n = 300)	
	Mean	Standard Deviation	Mean	Standard Deviation
Age (in years)	25.9	6.1	24.1	4.3
Age First Arrest (in years)	13.1	1.7	15.4	3.9
Education	11.2	1.9	12.1	2.2
Number of children	1.1	1.1	2.1	1.2
	Frequency	Percent	Frequency	Percent
Race				
Caucasian	136	45.3	110	36.7
African American	108	36.0	171	57.0
Asian American	3	1.0	0	0.0
Hispanic American	48	16.0	19	6.0
Other	5	1.7	1	0.3
Childhood Sexual Abuse				
Yes	126	42.0	210	70.0
No	174	58.1	85	28.3
Missing	5	0.8	5	1.7
Childhood Physical Abuse				
Yes	135	45.0	201	67.0
No	159	53.0	91	30.3
Missing	6	2.0	8	2.7
Current Sexual Abuse				
Yes	15	2.5	156	52.0
No	576	95.8	140	46.7
Missing	10	1.7	4	1.3
Current Physical Abuse				
Yes	12	4.0	144	48.0
No	277	92.3	151	50.3
Missing	12	4.0	5	1.7
Living with Conventional Partner				
Yes	213	71.0	184	61.3
No	87	29.0	116	38.7
Full-Time Employment (past 3 years)				
Yes	217	72.3	187	62.3
No	83	27.7	113	37.7
Gang Member				
Yes	198	66.0	133	44.3
No	102	34.0	167	55.7
Regularly Carry Weapon				
Yes	187	62.3	79	26.3
No	113	37.7	216	72.0
Missing			5	1.7
Sell Drugs				
Yes	192	64.0	176	58.7
No	93	31.0	118	39.3
Missing	15	5.0	6	2.0

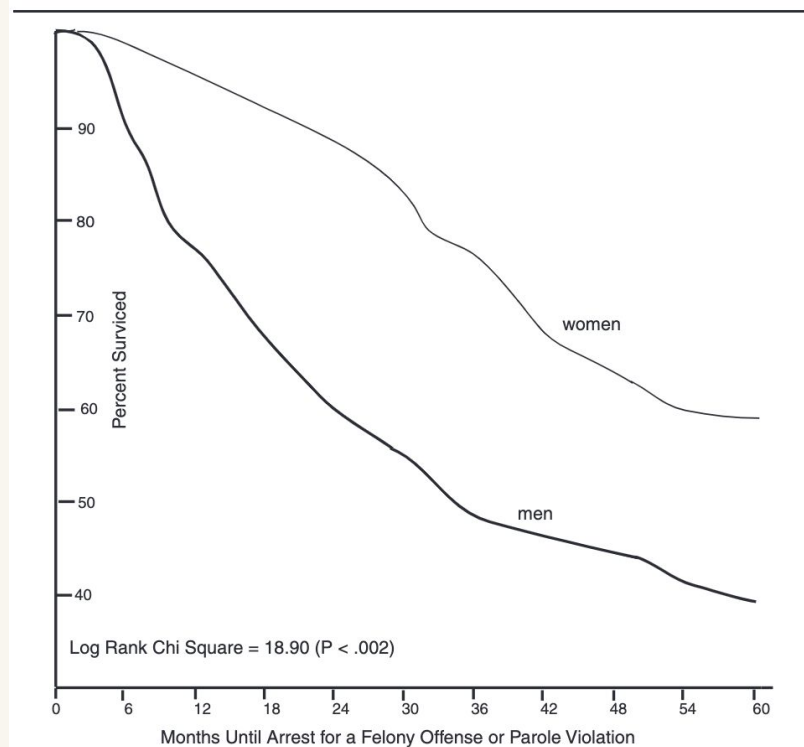


Figure 1 Survival Curves: Tenure in the Community